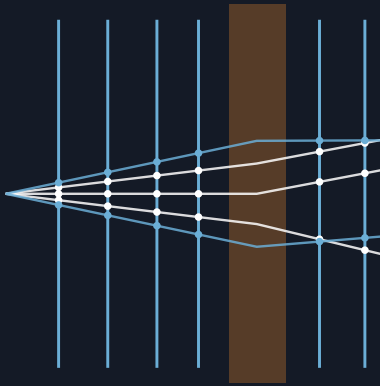
MUON BACKGROUND, DESIGN SPACE AND RECONSTRUCTION, LIVE

A veto for SHiP

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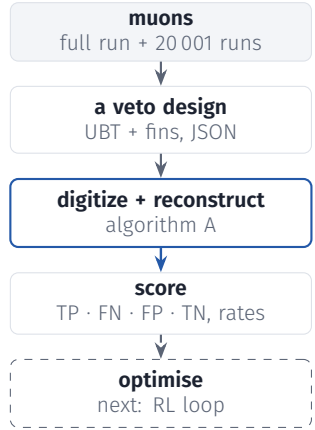
How to read this talk

Every plot with a slider is computed while you watch, by the code of the study (`veto/lib/veto`):

- the muon samples of the full simulation and the 20 001 old runs
- digitization, accidental hits and the veto reconstruction
- design comparisons, rerun on request

Slow widgets show a spinner and fill in; some show a quick answer first. 3D views rotate and zoom.

> The spill is **1 s** of $5 \cdot 10^{13}$ POT and the weights are muons per spill, so muons per spill and Hz are the same number.



PART 1

The problem

Muons from the target reach the detector; two of them can fake a decay in the vessel. The veto has to see them on their way in, without killing the signal.

SHiP and its muon background





Target at $z = 0$, **muon shield** (TRY6) to 30 m.

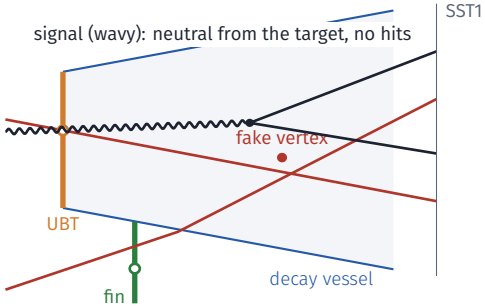
Decay vessel $z = 30.6$–80.6 m, 1×2.7 m entrance widening to 4×6 m.

UBT on the entrance window, **SST1** at 82 m, spectrometer magnet and timing detector behind.

Cavern wall at $x = -4.56$ m, floor at $y = -3.36$ m.

Muons per type (slider): bounced into SST1 · SST1 $p > 1$, timing · SST1 other · through an SBT wall · through the UBT window · bounced in the rock · straight past · stop in the rock · stop in the shield.

What the veto has to do



Two muons from the same spill, within ±200 ps, can form a vertex inside the vessel that points back to the target: a **fake signal**.

Each came in from outside: through the UBT window, the side walls, or after bouncing off the cavern. A hit where its track came from **explains** it.

muons per spill (simulated)	$1.8 \cdot 10^{10}$
crossing $z = 30.6$ m	$3.5 \cdot 10^{9}$
in the SST1 acceptance	$3 \cdot 10^{4}$
fake-vertex pairs, 15 years	**36** before any veto

> Veto every fake vertex, but do not veto the signal with **random hits** of the 10^{9}–10^{10} other muons.

PART 2

The data

Two simulations: 20 001 runs that keep only what reaches the spectrometer, and one run that keeps every muon.

Two simulations of the same muons

	20 001 old runs	**the full run** (`Full_dataset_z_3060_8200.pkl`)
muons	only those ending in the SST1 acceptance, ≈ 1 300 per run, 26.5 M rows	all 506 M simulated, one row each, 32 GB
states	start, z = 30.6 m, z = 82 m	z = 30.6 m and z = 82 m, or where the muon stopped
good for	pair background: ≈ $2 \cdot 10^5$ fake-vertex candidates	everything that misses the spectrometer: UBT and fin rates, veto-only muons
read with	a column store, 78 s to build	memory-mapped in place, a pass in ≈ 1 min

- The full run is a **rerun of seed 42**: same input muons, same states at 30.6 m; old file 42 holds its SST1 muons. Combined, it is not a 20 002nd run.
- `evt_n` = input-muon index in both; weights belong to the input muon (identical in both).
- Muons that bounce off the cavern differ between the two: bouncing is random from run to run.

Where the muons go

live:where_bar

70 % stop in the shield; 26 M range out in the rock behind the cavern wall.

125 M pass the vessel to z = 82 m outside the spectrometer: harmless themselves, but they light up any detector placed in their way.

Only ≈ 2 000 per run matter directly: they enter the vessel or reach SST1.

> The veto sees 10^9–10^{10} muons per spill to catch ≈ 10^4. Accidental hits are the real enemy.

Where the muons go, in Hz



The same muons as rates: muons per spill of $5 \cdot 10^{13}$ POT over the 1 s spill.

≈3.5 GHz cross z = 30.6 m and 2.7 GHz pass the vessel to z = 82 m.

≈29 kHz reach the SST1 acceptance (the three SST1 bars together), ≈27 kHz enter the vessel without reaching it.

> A veto detector sees GHz in total; the muons it must catch arrive at tens of kHz.

Bouncing off the cavern

live:bounce_view

live:bounce_map

live:bounce_ev

Grey dashed: straight line from the 30.6 m state; orange dotted: backward line from the 82 m state. They meet within 4 cm (median) at the bounce point, on the $-x$ wall or the floor. Right: all reconstructed bounce points trace the cavern.

The design-optimisation dataset

type	source	rows	muons / spill
SST1, bounced	old runs	6.0 M	$1.0 \cdot 10^4$
SST1, $p > 1$ GeV, timing	old runs	6.0 M	$8.2 \cdot 10^3$
SST1, other	old runs	14.5 M	$1.2 \cdot 10^4$
veto only	full run	975	$3.3 \cdot 10^4$
bounced in the rock	full run	575 k	$1.8 \cdot 10^7$
straight past	full run	15 M of 125 M	$2.7 \cdot 10^9$
stops in the rock	full run	15 M of 26 M	$7.7 \cdot 10^8$

`design-optimization-dataset-alpha-14-oct-2026`: 57 M muons, 5.8 GB.

Each type capped at 15 M rows and re-weighted, so `weight` is always muons per spill:

$$w = w_{\text{orig}} \cdot \frac{1}{N_{\text{runs}}} \cdot \frac{N_{\text{type}}}{N_{\text{kept}}}$$

Columns: states at 30.6 and 82 m, start state (old runs), bounce point, SBT walls crossed, type, run, `evt_n`.

> Veto-only and rock-bounced come from one run: right rate, few distinct muons. A new run will fix that.

PART 3

The design space

A UBT on the vessel entrance and thin fins reaching out from the side walls, left and right. Nothing on top or below.

UBT and fins











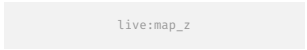




Sliders: UBT top · fins per side · gap · fin length · fin top [m]. Fin colour: $\log_{10}$ muons/m^2/spill, 2 (purple) to 9 (yellow). Black: drawn muons crossing nothing. Bars: geometric coverage.

Where the muons are







Muon density on a fin plane at z, every muon of the dataset; u is the distance from the vessel wall. Side: 0 right ($-x$), 1 left ($+x$). Dotted: cavern wall and floor limit.

A **lobe** at $y \approx 0$, 1.5–3 m from the wall, reaches 10^9 muons/m^2 per spill: the muons the shield sweeps past the vessel. Close to the wall and away from $y = 0$ it is 3–5 orders of magnitude quieter.

> A fin pays for every muon crossing it. The muons it is there for cross it near the wall.

Seven designs to learn from

		area
D0	UBT on the entrance only	3 m^2
D1	taller UBT, one 4 m fin per side at 30.6 m	56 m^2
D2	4 fins per side every 15 m, 3 m long, $y \pm$ 3 m	145 m^2
D3	7 fins on the right up to the wall, 3 short on the left	209 m^2
D4	8 fins per side, 5 m, y to 6 m: the stress test	624 m^2
D5	D2 with 50 cm or 5 cm cells	145 m^2

A design is a small JSON file: UBT edges and cell size; per fin its side, z, length, y range, cell size; detector efficiency and time resolution.

The loader applies the physical limits (floor, cavern wall, minimum UBT) and says what it changed.

All: 10 cm cells, $\varepsilon = 0.99$, $\sigma_t = 0.5$ ns, no tilt yet.

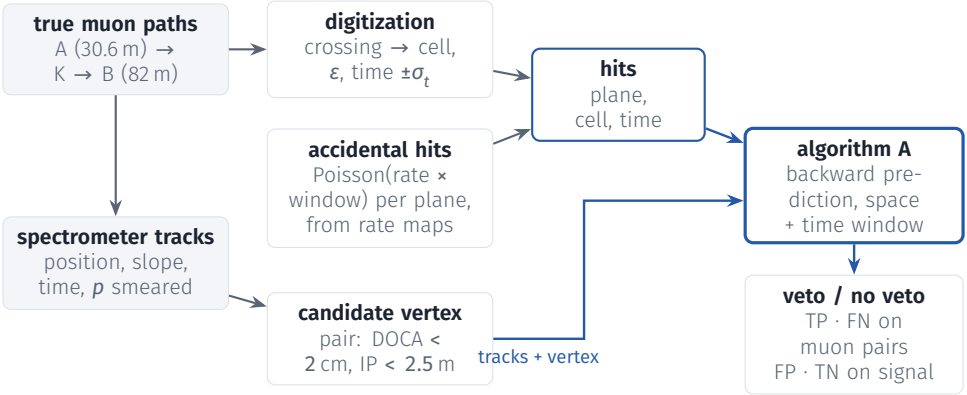
> In the RL environment fins are placed one after another along z with a minimum gap.

PART 4

Reconstruction

Events are short time windows around a spectrometer candidate. Each spectrometer track is followed back to the veto planes; a hit where it should be explains it.

The chain



Signal events: a long-lived neutral from the target decaying in the vessel into two tracks, with the pairs' selection; no own veto hits, only accidental hits can veto them.

Events are time windows, not spills





cell size [cm]; 1 s spill

A fin at 30.6 m sees ~ 10^9 muons per second: simulating whole spills hit by hit is hopeless.

Instead, around each candidate: the own muons' hits, plus per plane

$$n_{\text{acc}} \sim \text{Poisson}\left(R_{\text{plane}} \cdot \frac{2W}{T_{\text{spill}}}\right)$$

random hits, placed according to the full-statistics rate map and digitized like real ones.

Blue box: a search window near the wall. What matters is the chance a random hit lands in it: rate density × box × window.

Orange squares: random hits of one ±1.5 ns window on the right fin of D2, at the chosen cell size.

The detector and spectrometer models

Veto planes (UBT and every fin, the same model)

- one layer of cells at fixed z (10 cm default)
- a crossing makes one hit with probability ε (0.99)
- hit = cell centre + time, σ_t = 0.5 ns, muons at c
- no direction, no energy, no noise, no dead time (yet)
- a muon's own crossing comes from its true path, bounce included

Spectrometer (track at z = 82 m)

- position ±1 mm, time ±0.1 ns
- slope ±0.1 mrad ⊕ 2 mrad·(1 GeV/p)
- momentum as true, used only to size the window

Direction comes only from the spectrometer. The UBT gives no usable direction; neither do the fins.

Candidates

- background: opposite-charge muon pairs of the same old run, p > 1 GeV, in the timing acceptance, DOCA < 2 cm, vertex in the vessel, IP < 2.5 m: **196 736** pairs
- each weighted by $w_1 w_2 N_{\text{runs}} \cdot 2\Delta t / T_{\text{spill}}$, Δt = 200 ps
- signal: long-lived neutrals from the target, nearly along the beam ($\theta \approx p_T/p$), decaying in the vessel into two tracks; same selection as the pairs
- single SST1 muons (200 k) for per-muon efficiencies

The signal: from the target, not from Mars



A long-lived neutral from the **target**: p = 10–100 GeV (log-uniform), p_T ~ exp(1 GeV), so $\theta \approx p_T/p$.

Decays uniformly in z in the vessel, m = 1 GeV → two muon-mass tracks.

The pairs' selection: SST1, p > 1 GeV, timing, IP < 2.5 m. **37 %** pass; median daughter angle 22 mrad.

Orange: its tracks followed back to z = 30.6 m; grey: muon density.

> Two thirds cross the quiet UBT; the rest hit the fins next to the wall, where random hits veto them.

Algorithm A: follow every track back

For each spectrometer track:

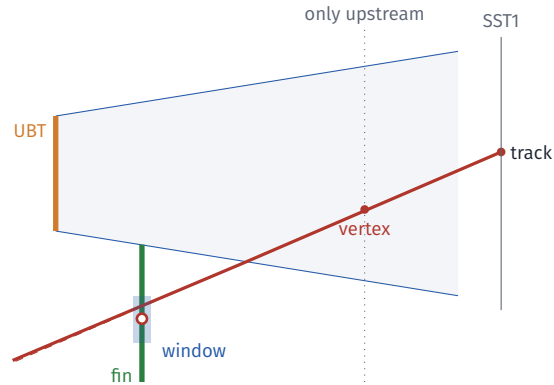
1. extrapolate backward in a straight line from $z = 82$ m (no field in the vessel)
2. on every veto plane **upstream of the vertex**: predicted cell (u, y) and time $t - \ell/c$
3. look for a hit within

$$|\Delta u|, |\Delta y| \le r_0 + \left(a + \frac{b}{p}\right) d + \tfrac{\text{cell}}{2}, \quad |\Delta t| \le n\,\sigma$$

4. a hit found: the track is **explained**

A candidate is vetoed if either track is explained.

> No combinatorics: one binary search per (event, plane) and a scan of that plane's few hits.



The spatial window

live:window_plot

live:r_ms

$r = r_0 + (a + b/p)\,d$ with $r_0 = 5$ cm, $a = 0.5$ mrad and b the slider, in mrad·GeV: the spectrometer's slope error grows with the lever arm d, multiple scattering with $1/p$.

Title: how many of the true crossings of straight SST1 muons (D2, 60 k muons) fall inside the window, computed now.

> For straight muons the prediction is within millimetres: the window can be tight. A bigger window only buys more random hits.

The time window

live:time_plot

live:time_trade

live:sig_t

live:n_sig

The predicted time includes the time of flight from the plane to z = 82 m. The window is $\pm n\sigma$ with $\sigma = \sigma_t^{\text{veto}} \oplus \sigma_t^{\text{track}}$: wider loses fewer own hits, but lets in proportionally more random hits.

Right: signal loss scaled from D2's 1.3 % at 3σ and 0.5 ns, since random hits grow with the window.

Algorithm A on real events



design: 0 D0 · 1 D1 · 2 D2 · 3 D3 · 4 D4 · 5 D5 coarse · 6 D5 fine. kind: 0 pair vetoed · 1 pair missed · 2 signal lost to a random hit · 3 bounced muon missed.

Solid: true path; dotted: backward prediction; white: own hits; diamond: the hit that explained the track; grey: random hits.

The same event from above



x–*z* view. Thick lines: UBT (orange) and fins (green). Blue bars: the search window on each plane the track is followed to. Same sliders as the previous slide.

What it costs

design	planes	hits / event	comparisons	µs / event
D0	1	0.1	40 k	0.5
D1	3	16	3.2 M	4.8
D2	9	39	4.0 M	12.5
D3	11	49	4.9 M	16.7
D4	17	90	7.2 M	33.3

196 736 background events; numpy, one core of a laptop.

A full evaluation of a design (196 k pairs, 100 k signal, 200 k single muons) takes **≈6 s** on one core, linear in hits per event.

Nothing to cap: algorithm A never combines hits.

> On a GPU, or with the factorised estimate, 10–100 ms per design: fast enough for an RL loop.

PART 5

Results

The background is easy to veto. Keeping the signal, and catching the last muon pairs, is where designs differ.

RESULTS

Seven designs

live:compare

	D0	D1	D2	D3	D4	D5 coarse	D5 fine
background vetoed	25 %	99.992 %	99.997 %	99.992 %	100.000 %	99.997 %	99.997 %
pairs surviving	171 020	54	27	19	6	24	27
signal lost	0	1.13 %	1.28 %	1.49 %	1.76 %	4.72 %	1.00 %

RESULTS

Turn the knobs











design: 0 D0 · 1 D1 · 2 D2 · 3 D3 · 4 D4 · 5 D5 coarse · 6 D5 fine. The whole chain reruns: a quick answer on 15 % of the events, then all of them (≈5–10 s).

Figure of merit

live:fom

live:fom_design

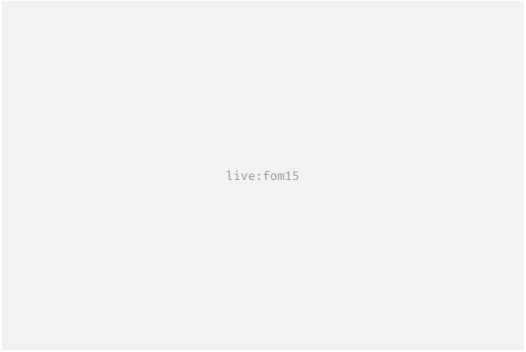
Positive = vetoed. Background: events in 15 years; signal: fractions.

design	TPR	FNR	FPR	TNR	B / 15 y	Punzi
D0	25.2 %	74.8 %	0	100 %	27	0.15
D1	99.992 %	0.008 %	1.13 %	98.87 %	0.003	0.637
D2	99.997 %	0.003 %	1.28 %	98.72 %	0.001	0.645
D3	99.992 %	0.008 %	1.49 %	98.51 %	0.003	0.634
D4	99.9998 %	0.0002 %	1.76 %	98.24 %	0.0001	0.651
D5 coarse	99.997 %	0.003 %	4.72 %	95.28 %	0.001	0.623
D5 fine	99.997 %	0.003 %	1.00 %	99.00 %	0.001	0.647

Punzi = $\varepsilon_S/(1.5 + \sqrt{B})$, ε_S = TNR. Every design with fins has $B \ll 1$, so the FoM is set by the signal efficiency.

> B rests on a few dozen surviving pairs and some muons weigh 100× more: one pair can move it by 5×. The factorised estimate comes next.

Figure of merit: the 15-year run







The same confusion matrix, as **counts over 15 years**: $6 \cdot 10^{20}$ POT in 1 s spills of $5 \cdot 10^{13}$ POT, $1.2 \cdot 10^{7}$ spills.

Fake vertices: 36 muon pairs over the whole run look like a decay in the vessel. The veto removes all but a tiny fraction: FN is the expected number of **fake discoveries**.

Signal: its total is model dependent; the slider sets how many selected decays SHiP would see. FP are lost to random hits, TN kept.

$P(\geq 1 \text{ fake}) = 1 - e^{-\text{FN}}$: the chance the whole run has even one.

> Background-free means FN $\ll$ 1: true for every design with fins. They differ in how much signal they keep.

RESULTS

Experiments

variation of D2	FPR	TPR	B / 15 y
as is	1.21 %	99.997 %	0.001
σ_t 0.5 → **0.1 ns**	**0.41 %**	99.98 %	0.006
σ_t 1 ns	2.19 %	99.997 %	0.001
cells 10 → 3 cm	0.83 %	99.997 %	0.001
σ_t 0.1 ns **and** 5 cm cells	**0.30 %**	99.98 %	0.006
time window 3 → 2σ	0.89 %	99.91 %	0.03
time window 4σ	1.65 %	99.998 %	0.0008
veto needs hits on 2 planes	0.002 %	36 %	23
without the fins at 30.6 m	0.13 %	61 %	14
fins only 1 m long	0.20 %	98.9 %	0.4
fins only y = −1..1 m	1.21 %	94.2 %	2.1
D4, σ_t 0.1 ns and 5 cm cells	0.49 %	99.9998 %	0.0001

60 k signal decays per row; `bin/run_reco.py` on one core.

Time resolution is the biggest lever on signal loss, then cell size: 3–4× less, at no cost in background.

The **fins at 30.6 m** veto most of the background and cause most of the signal loss.

A **2-plane rule** fails: most muons cross exactly one plane. Redundancy has to be built into the geometry.

> Short fins trade 0.2 % signal loss for about 0.4 background events: the kind of choice the optimiser has to make.

RESULTS

Why the last pairs survive





A pair survives only if **neither** muon is explained. For a straight muon crossing a single plane that is one chance:

$$P_{\text{miss}} \approx (1 - \varepsilon) + 2\,\Phi(-n) \approx 1.3\,\%$$

so a pair escapes with $\approx 2 \cdot 10^{-4}$: ≈ 30–40 of 196 k.

Bounced muons are followed back only after their bounce: the prediction misses the planes they crossed before it by metres.

> Redundancy: two planes on every incoming path square the miss probability.

Bounced muons, in perspective

background pairs	count	weight
no bounced muon	193 873	98.7 %
one bounced muon	2 863	1.3 %
two bounced muons	0	0 %

Single SST1 muons vetoed by algorithm A:

	D2	D3	D4
$p > 1$ GeV, timing	99.1 %	99.2 %	99.3 %
other	99.3 %	99.7 %	100.0 %
bounced	74 %	87 %	88 %

Bouncing costs most of the momentum (median 3 → 0.3 GeV), so bounced muons rarely pass $p > 1$ GeV: 23 % of SST1 muons, 1.5 % of the pairs.

For single muons algorithm A's blind spot is real; for the pair background it is a small effect, and the partner muon usually explains the pair.

> A hit-driven algorithm B could use the hits before the bounce: tracklets in the fins pointing at the vessel.

Granularity: the signal-loss knob

live:cell_scan

live:cs_design

Same geometry, cell size from 3 to 50 cm, 30 k signal events per point.

Smaller cells shrink the search box, so fewer random hits fall in it: signal loss drops from 4.7 % (50 cm) to 1.0 % (5 cm) for D2.

But the busiest 10 cm cell already sees **60 MHz**, the UBT 400 Hz. Dead time and pile-up are not modelled yet and will matter in the lobe.

> Detector technology near $|x| \approx 2$ m is a design question of its own.

PART 6

Next

What the reconstruction still ignores, and how the design loop will use it.

NEXT

What is not in yet

Reconstruction

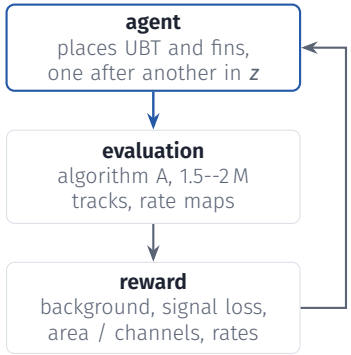
- algorithm B: hit-driven tracklets across fins (sees bounces), with a time cap per event
- factorised background: $\sum w P^{(1)}_{\text{miss}} P^{(2)}_{\text{miss}}$ instead of counting a handful of survivors
- dead time and pile-up at MHz rates per cell
- tilted fins (geometry and hits exist, not in the reconstruction)

Physics and data

- muon DIS in the walls and fins, neutrinos
- muons from the same proton collision (correlated in time)
- specific signal models (masses, decay modes, lifetimes) beyond the generic two-body decay
- more runs of the veto-only muons: per seed, everything crossing the vessel walls or UBT window, plus a 10^{-5} sample of UBT-plane crossers

NEXT

Towards the optimisation loop



Per design: ≈1 M SST1 muons, $2 \cdot 10^5$ pairs, 10^5 signal decays, and **no** flux muons as particles: they enter as rate maps.

Accidental vetoes analytic, $1 - e^{-\sum \text{rate} \times \text{window}}$: no Monte-Carlo noise in the reward.

> Expect the agent to cover incoming paths twice near the wall and to stay out of the lobe.

NEXT

Open questions

- POT per spill: are the weights muons per 1 s spill of $5 \cdot 10^{13}$ POT?
- The repeated-`evt_n` weight correction of the original analysis: the dataset uses the plain average over runs.
- Cavern geometry in the simulation: muons cross $x = -4.56$ m and reach $x \approx -9$ m.
- Veto technology near the vessel: rates per cell, time resolution, dead time.
- The new run: what to store per seed (above), and keep the start state and `evt_n`.

NEXT

Summary

Data: the full run and 20 001 old runs, combined into a 57 M-muon design dataset with per-spill weights; cavern bounces reconstructed to 4 cm.

Design space: a UBT plus fins from the side walls; seven designs from minimal to dense.

Reconstruction: time-window events with accidental hits from full-statistics rate maps; a track-driven veto, ≈10 µs per event, no combinatorics.

> Background is vetoed at > 99.99 % by any design with fins ($B \ll 1$ in 15 years). The figure of merit is then the signal efficiency: 98.7 % for D2, 99.7 % with 0.1 ns timing and 5 cm cells.